

Python Algorithm Learning

Project Submission Report

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Official VIT Bhopal logo was not available in the supplied project, and a reliable official image source could not be confirmed. No substitute logo has been used.

Index

Introduction	3
Problem Statement	3
Objectives	3
Functional Requirements	4
System Architecture	5
Workflow	6
Module Description	7
Testing and Results	8
Challenges Faced	9
Future Enhancements	9
Conclusion	10

Introduction

Python Algorithm Learning is a small terminal program for students learning introductory algorithms. A student selects one of four categories, runs an example, or answers short practice questions. The program checks answers and shows session results.

Problem Statement

Algorithm examples are often scattered across separate files. Beginners need a straightforward way to find an example, try it, and check their understanding.

Objectives

1. Organize algorithms into four categories.
2. Handle common invalid inputs clearly.
3. Add topic and mixed practice with answer checking.
4. Show attempts, correct and incorrect answers, and accuracy for the current module session.

Functional Requirements

The root menu opens Number Theory, Array Algorithms, Sequences, or Data Conversion. Each module has its own menu and can return to the root menu with option 0.

Users can run the listed algorithms, enter values through prompts, and receive results. Invalid numbers and invalid binary strings are handled with readable messages.

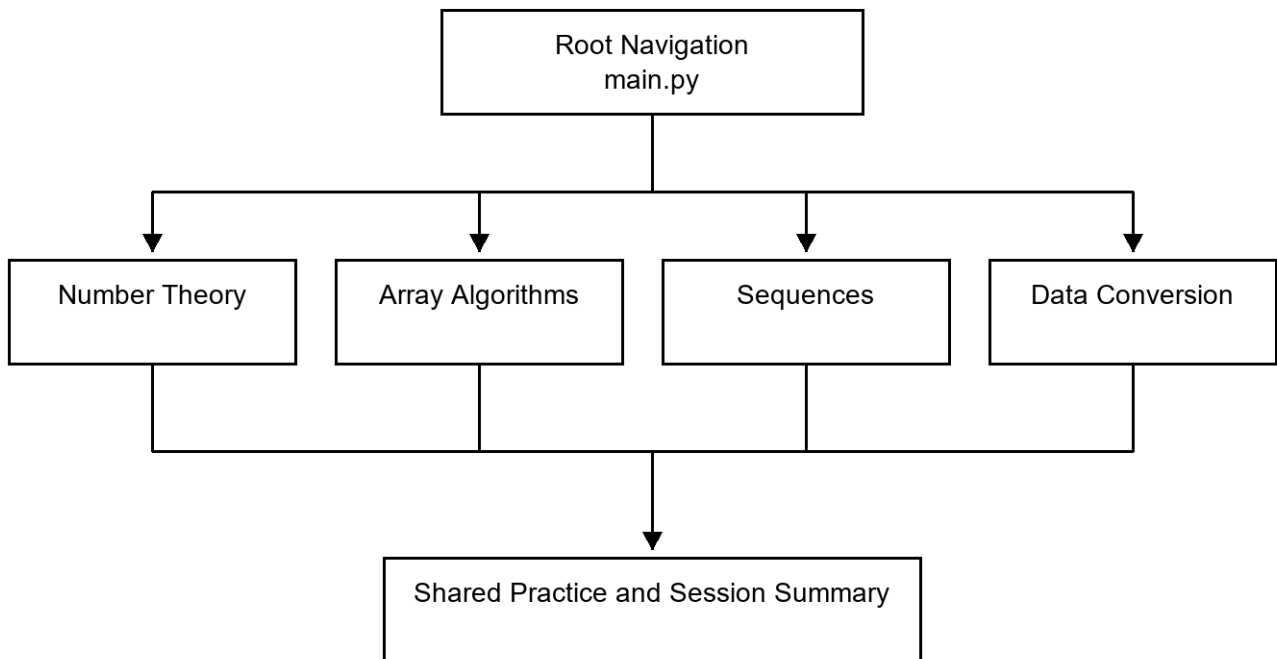
Topic practice uses that module's question list. Mixed practice randomly selects from questions across the four categories. Users choose a question count from 1 to 20. Results are kept in memory while the module stays open.

Practice answers are compared with the expected text answer. Accuracy is the correct answer count divided by attempts. Results are not saved after the program exits.

System Architecture

The root program launches the selected module. The module can run its algorithms or call the shared practice code. It returns to the root menu when the user selects 0.

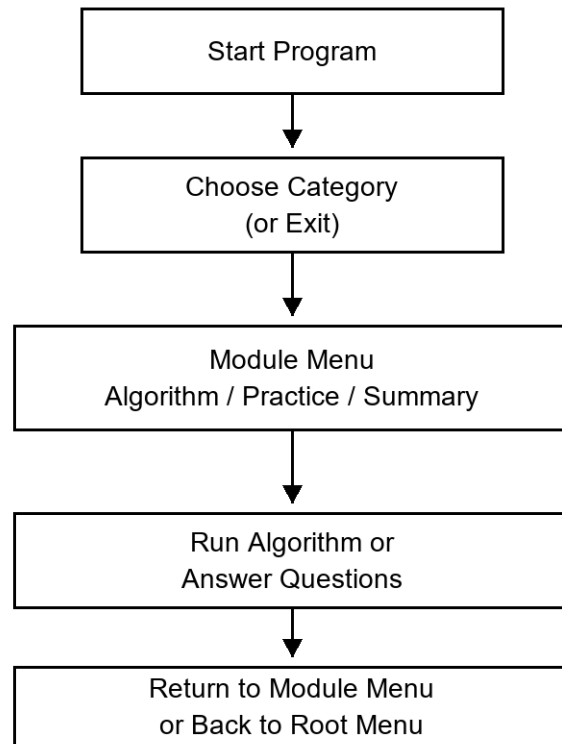
PYTHON ALGORITHM LEARNING - SYSTEM ARCHITECTURE



Workflow

The program repeats the menu flow until the learner exits. Practice questions are selected randomly and checked as they are answered.

PYTHON ALGORITHM LEARNING - WORKFLOW



Module Description

Number Theory: GCD, prime checking, prime factorization, and LCM.

Array Algorithms: array reversal, counting occurrences, finding the kth smallest value, linear search, and bubble sort.

Sequences: Fibonacci terms, factorial, and conversion of space-separated text into an integer list.

Data Conversion: signed decimal integers to hexadecimal and binary numbers to decimal.

Practice: topic and mixed question banks, answer checking, and an in-memory summary.

Testing and Results

Automated checks were run with Python unittest. They cover number theory results, array algorithms, sequence boundary cases, invalid conversion and sequence inputs, and practice scoring.

Representative results:

- $\text{GCD}(54, 24) = 6$; $\text{LCM}(4, 6) = 12$.
- $\text{Fibonacci}(7) = [0, 1, 1, 2, 3, 5, 8]$; $5! = 120$.
- Decimal 255 converts to FF; binary 1010 converts to 10.
- Practice scoring records 2 attempts, 1 correct, and 1 incorrect in the unit check.

Full-screen application screenshots could not be captured: no desktop apps or browser were exposed to the available capture interface. See screenshots/TEST_OUTPUT.txt for supplementary console test evidence.

Challenges Faced

The original root menu had incomplete control flow, and the Data Conversion entry point contained an array menu. Separate module menus and a root launcher corrected those navigation issues.

The original module copies contained import-time prompts and inconsistent input behavior. The completed menus use local functions and explicit validation for the requested algorithms.

Future Enhancements

Possible improvements include exporting student performance reports in PDF and text formats, adding more algorithms and practice questions, and offering more flexible mixed tests and question randomization.

A future AI learning agent could find online learning resources and organize lessons by selected topic and learning preference. It could offer structured video links, AI-generated notes, or highlighted notes from suitable materials. This is not part of the current program.

Conclusion

Python Algorithm Learning brings four groups of beginner algorithms together in one menu-driven program. Learners can run examples, practise topic and mixed questions, and review current-session performance. The project provides a base for future saved reports and expanded learning resources.